

ChristosHadjinikolis

Engineering Manager | Production ML & Streaming Systems

Ph.D. Computer Science | UCL Research Associate | AI Standards Expert @ ISO/CEN-CENELEC JTC'21

✉ christos.hadjinikolis@gmail.com

mediums



about me

Engineering Manager for ML and real-time data systems with **16 years** across AI research, consulting, backend engineering, and live data platforms. At Vortexa, own a major **live streaming backend estate**, lead a **10-person cross-functional team**, and drove Data Production growth from **4 to 30+ people**. I build teams, operating models, and platforms that turn noisy data and experimental models into dependable, client-facing ML systems. UCL/JTC'21 work keeps auditability, explainability, versioning, and safe AI adoption close to my engineering practice.

work-experience

Vortexa Ltd, London, UK 12/2020–present

Engineering Manager | ML Systems Lead

Own **engineering strategy and delivery** for Vortexa's **live streaming intelligence estate**: a client-facing backend platform turning roughly **6M vessel-position records/hour** into intelligence for **13.5K monitored vessels**. Lead **live-processing strategy** with a **10-person cross-functional team** (**6 MLE/DE/DS, 2 Product, 2 SMEs**); reliability, explainability, monitoring, and response readiness are delivery requirements.

- **People & Org Scaling**: Directly manage the MLE/DE/DS core and lead a **10-person delivery team**; drove Data Production from **4 to 30+**; hiring manager for **6 roles (31+ candidates, 60+ interviews overall)**; retained/progressed the full current team.
- **Streaming Platform**: Led the two-year **Kafka Streams-to-Flink migration** for critical live pipelines, moving toward dataflow-based scaling, independent state/checkpointing, stronger operational visibility, and lower maintenance effort.
- **Production ML**: Established Vortexa's research-to-production ML lifecycle: **MLflow**, model/data versioning, service deployment, replay/evaluation, live accuracy monitoring, explainability, and auditability.
- **Reliability**: Built monitoring, runbook, rollback/fallback, Jira-alert, and shared on-call practices, keeping **MTTR under 30 min** for a client-visible live estate.
- **Operating Model**: Turn product-semantic friction into measurable workstreams; scale delivery through repo archetypes, ADRs, dev containers, Backstage, local E2E tests, DORA/Jira signals, and an architecture forum. Cut a **3h DAG loop to under 5 min**.

DataReply, London, UK 04/2016–12/2020

Senior Consultant | Data Scientist / ML Engineer

Joined Data Reply's London spin-off as its first consultant and helped grow it from **3 practitioners and 2 managers to 30+ consultants**. Progressed to Senior Consultant, leading client meetings, scoping, consultant placement, hiring/mentoring, and ML/data delivery across UBS, Vodafone, CNHi, and other enterprise clients. Ran the London Flink Meetup for five years.

- **Client Delivery**: Drove stakeholder agendas, scoped goals, facilitated delivery, and supported hiring/placement across banking, telecoms, manufacturing, publishing, and gaming.
- **Vodafone Group**: Led cloud/data delivery; worked on Infinity, a GCP/**Kubeflow** DS platform, and Red Agent, a mobile-network feature-engineering framework.
- **CNHi**: Lead Data Scientist/Scrum Master for predictive maintenance, guiding a DS team using **PySpark** over live vehicle sensory data.
- **UBS**: Grew from Data Scientist into ML Engineer through XP/pairing; built graph/process-mining analytics, observability, and **Kafka/Elasticsearch/Python** real-time pipelines.

Roles in UK Institutions 2010–2016

Coding Teacher @DGC | Teaching Assistant @KCL/UCL

Built the communication and presentation discipline I now rely on as a leader by teaching Java/Python and core CS at David Game College while completing doctoral research; Teaching Assistant at KCL and UCL.

organisation

org scale-up
hiring loops
mentoring/retention
cross-functional delivery

systems

client-facing ML
high-volume streaming
observability
reliability

engineering

operating models
streaming platforms
DORA/Jira signals
local E2E tests

technical

ML systems
Java/Kotlin
Python/SQL
Kafka/Flink
Airflow/AWS/GCP

public

JTC'21 standards
UCL research
PyPI: dynamicio
skeleton-replay
JetBrains plugin

education

2010–2014	Ph.D. in Computer Science <i>Thesis: Persuasion Dialogues & Opponent Modelling.</i> Developed logical inference methods for large Knowledge Graphs , combining symbolic reasoning, graph analytics , and Bayesian techniques. Research contributed to scalable AI grounded in formal semantics. Best Poster Award , IJCAI 2013 (413 submissions); EPSRC PhD Scholarship ; Outstanding TA Award ; Graduate Certificate in Academic Practice . Access Publications on Google Scholar	King's College London, UK
2004–2010	Diploma (BEng) in Computer Engineering Five-year polytechnic degree in Computer Science and Engineering; strong mathematics focus (10 courses), theoretical CS, networking, telecommunications. <i>Majored in Artificial Intelligence</i> .	University of Thessaly, GR
2002–2004	Military Officer: Second Lieutenant Completed national service as a commissioned Second Lieutenant.	National Guard, CY

committees/affiliations

since 10/2024	University College London, Dept. of Information Studies Research/teach practical AI standardisation: AI Act, auditability, versioning, explainability, safe adoption.	Associate Researcher
since 01/2021	ISO - International Organisation for Standardisation Member of JTC'21 Working Group 3, developing AI standards aligned with EU policy and international norms.	Committee Expert Member for JTC'21

selected public tooling

PyPI	dynamicio Free tooling for explicit I/O seams, schema/data validation, and local/dev/prod dataset switching so ML pipeline logic can be tested faster.	ML/data I/O contracts
PyPI	skeleton-replay Python tool turning script/pytest runs into traces, architecture snapshots, workflow evidence, and replayable reports for review, debugging, onboarding, and LLM-assisted understanding.	Runtime architecture evidence
JetBrains	Skeleton Replay plugin Free PyCharm/IntelliJ plugin bringing Skeleton reports, workflow evidence, and source navigation into the IDE.	IDE workflow

talks/interviews

2023	Bill Raymonds, Agile in Action Agile Data Science: The Agile Journey at Vortexa, Details: YouTube	Podcast Interview
2022	Open Data Science Conference dynamicio: PyPI library for abstracting I/O in ML/data workflows, Details: here	Industry Conference Talk
2020	iunera: Big Data Innovators The Agile approach in data science, Details: here	Interview Blog Post
2020	Big Data Warsaw Monitoring & Analysing Communication and Trade Events as Graphs, Details: here	Industry Conference Talk
2018	Connected Data London AI tribes and interconnections: what's graph got to do with it?, Details: here	Panelist @ Industry Conference
2018	Minds Mastering Machines A talk on doing Data Science the Agile Way, Details: here	Industry Conference Talk

selected certificates & awards

07/2020	AWS Certified Machine Learning - Specialty Professional certification	AWS
01/2019	Process Mining Course certificate	Coursera Course Certificates
10/2017	Google Certified Professional - Data Engineer Professional certification	Google Cloud Certified
07/2017	Advanced Elasticsearch: Data modelling Professional training	Elastic
06/2016	Graph Analytics for Big Data Course certificate	Coursera Course Certificates
05/2016	Apache Spark Developer Certification Developer certification	O'Reilly Media
04/2016	Neo4j Certified Professional Professional certification	Neo Technology, Creators of Neo4j